SAKSHAM PANDEY

Full Stack Engineer | React • TypeScript • Python (FastAPI) • Node.js • Distributed Systems

Patna, India • +91 95467 48850 • pandeysaksham180@gmail.com
linkedin.com/in/saksham-pandey-184515250 • github.com/sakshampandey380

PROFESSIONAL SUMMARY

High-impact Full Stack Engineer with 3+ years of production experience architecting scalable distributed web systems, real-time protocols (WebRTC, WebSocket), and AI-integrated backend microservices. Proven track record building high-concurrency client/server architectures using React, TypeScript, FastAPI, Node.js, PostgreSQL, and Redis. Delivered applications with 99.8% uptime, sub-150ms latency, and 76% query performance improvements. Currently pursuing Integrated BS-MS in Artificial Intelligence and Cybersecurity at IIT Patna, focused on low-latency state management, robust database indexing, and automated CI/CD containerization.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript (ES6+), Python, SQL, Go (Basics), HTML5, CSS3

Frontend: React.js, Next.js, Redux Toolkit, Context API, Tailwind CSS, WebRTC, WebSocket, Responsive UI

Backend: FastAPI, Node.js, Express.js, Flask, RESTful APIs, Microservices, OAuth 2.0, JWT Authentication, Redis Caching

Databases: PostgreSQL, MySQL, Redis, Database Schema Design, SQL Indexing & Query Optimization

DevOps & Tools: Docker, Git, GitHub Actions, CI/CD Pipelines, Linux/Bash, Vercel, Render, Postman, Agile

PROFESSIONAL EXPERIENCE

Independent Full Stack Engineer — Contract & Client Projects

Jan 2023 – Present | Remote

- Architected and delivered 6+ production-ready full-stack applications using React, TypeScript, FastAPI, Node.js, and PostgreSQL, consistently maintaining **99.8% uptime** across client deployments.
- Engineered role-based access control (RBAC) and OAuth 2.0 / JWT authentication systems, mitigating XSS and CSRF vulnerabilities across high-volume admin portals.
- Refactored legacy REST APIs and implemented composite SQL indexing on PostgreSQL/MySQL databases, reducing average query response latency from **280ms to 65ms (76% improvement)**.
- Containerized full-stack microservices using Docker and established automated CI/CD deployment pipelines with GitHub Actions on Vercel and Render, accelerating product iteration velocity by **40%**.

KEY ENGINEERING PROJECTS

Viora Meet – Offline-Aware Video Conferencing Engine

React • FastAPI • WebRTC • WebSocket | github.com/sakshampandey380/viora_google-meet_copy

- Built high-performance video platform delivering **sub-150ms peer-to-peer latency** via WebRTC data channels and asynchronous FastAPI WebSocket signaling layer.
- Devised custom local-network (LAN) fallback mechanism enabling seamless multi-peer audio/video streaming during wide-area internet blackouts.
- Engineered modular screen sharing, canvas-based streaming, real-time mesh messaging, and automated reconnection recovery handlers.

ZipHire – AI-Powered ATS Resume Parsing Engine

React • Python (FastAPI) • PostgreSQL • NLP | github.com/sakshampandey380/resumeai_pro

- Engineered automated ATS parsing engine processing multi-format documents (PDF/DOCX) using NLP token extraction and structural AST analysis.
- Implemented vectorized keyword-matching algorithms evaluating candidates against job descriptions with **94% semantic accuracy**.
- Designed analytics dashboard with interactive scoring telemetry, secure credential management, and role-based multi-tenant isolation.

NexusChat – Distributed Real-Time Messaging Architecture

React • FastAPI • Redis • MySQL | github.com/sakshampandey380/Chat_App

- Constructed end-to-end event-driven messaging service with persistent bi-directional channels, dynamic user presence tracking, and delivery receipts.
- Optimized conversation history using paginated SQL cursors and Redis caching, preserving zero UI-stutters over **100k+ historical message threads**.

EDUCATION

Indian Institute of Technology Patna

Integrated BS-MS in Artificial Intelligence and Cybersecurity | 2025 – Present (Expected 2030)

Coursework: Data Structures & Algorithms, Distributed Systems, Network Security, Cryptography, Low-Latency Systems

CERTIFICATIONS & HIGHLIGHTS

- HackerRank: Gold Badge in Python & Problem Solving
- Advanced Full-Stack Engineering: REST APIs & Microservices
- System Design: Scalable Microservices, Dockerized CI/CD Pipelines
- Languages: English (Proficient), Hindi (Native), Russian (Conversational)